
1. INTRODUCTION TO QUANTUM STATES

1.1 Quantum Notation Basics

Before we talk about neutrinos, we need to learn some quantum mechanics notation!

States and Kets: In quantum mechanics, we describe the state of a particle using special brackets called kets that look like this: $|\text{something}\rangle$

Examples:

- $|\uparrow\rangle$ means spin up state
- $|\downarrow\rangle$ means spin down state
- $|\nu_e\rangle$ means electron neutrino state
- $|\nu_\mu\rangle$ means muon neutrino state

This notation was invented by physicist Paul Dirac and is called Dirac notation or bra-ket notation.

Classical vs Quantum: In everyday life, things are in definite states. A coin is either heads or tails. A light switch is either on or off. But in quantum mechanics, particles can exist in multiple states simultaneously until we measure them! This is called **superposition**.

Writing Superpositions: We write a superposition by adding states together with numbers in front:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Here, a and b are numbers called **probability amplitudes**. They tell us “how much” of each state is present. This entire equation tells me the state of a quantum system: We call this a wavefunction.

Example: An electron's spin state might be: $|\text{Spin}\rangle = 0,316|\uparrow\rangle + 0,948|\downarrow\rangle$

This means the electron is in a superposition of spin up and spin down!

The numbers in front of states (probability amplitudes) aren't directly the probabilities. We need to square them first!

The Rule: To find the probability of measuring a particular state, square the amplitude:

$$\text{Probability} = (\text{amplitude})^2$$

Example: For $|\text{Spin}\rangle = 0,316|\uparrow\rangle + 0,948|\downarrow\rangle$:

- Probability of measuring spin up: $(0,316)^2 = 0,100 = 10\%$
- Probability of measuring spin down: $(0,948)^2 = 0,899 \approx 90\%$

Notice that $10\% + 90\% = 100\%$. The probabilities must always add up to 1 (or 100%). This is called **normalization**, and it means we're guaranteed to find the particle in *some* state when we measure it!

Important: Before measurement, the particle is truly in both states at once. The act of measuring collapses the superposition to just one outcome. This is what makes quantum mechanics so strange and different from everyday experience!

1. A particle is in the state: $|\psi\rangle = 0,6|A\rangle + 0,8|B\rangle$

- a) What is the probability of measuring it in state A?
- b) What is the probability of measuring it in state B?
- c) Check that your probabilities add up to 1.

2. A coin is tossed up into the air. What are the possible states? What is the probability of heads? Of tails? Can you write a wavefunction to describe its state?

2. NEUTRINOS

2.1 Neutrino Oscillations

Now let's apply what we learned to real particles: neutrinos!

Neutrinos exist in three flavor states: Electron neutrinos ($|\nu_e\rangle$), Muon neutrinos ($|\nu_\mu\rangle$), and Tau neutrinos ($|\nu_\tau\rangle$). They are produced in various processes, such as nuclear fusion in stars, beta decay, and cosmic rays interacting with the atmosphere.

We learned that neutrinos are not fixed in one flavor state; they oscillate between flavors as they travel! This phenomenon occurs because neutrinos are created and detected as flavor states, but they travel as mass states (quantum superpositions of flavor states).

As a neutrino travels, the probability of finding it in different flavor states changes over distance.

$$P(\nu_e \rightarrow \nu_\mu) = \sin^2(2\theta) \sin^2\left(\frac{1.27 \Delta m^2 (eV^2) L(km)}{E(GeV)}\right)$$

Equation 1: Neutrino Oscillation Probability

Parameters:

θ : Mixing angle (describes how much the flavor and mass states are mixed)

Δm^2 : Mass-squared difference between the two mass eigenstates

L: Distance traveled by the neutrino

E: Neutrino energy

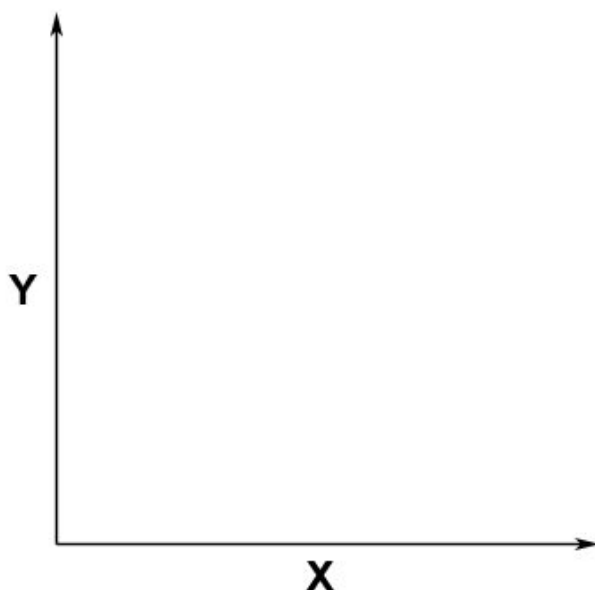


Figure 1: Probability vs Distance

3. Choose different values of L (try L = 100 km, 500 km, 1000 km, 1500 km, 2000 km) and plot the probability that an electron neutrino has changed into a muon neutrino. Here, we assume $E = 1$ GeV, the mass squared term is 7.5×10^{-5} , and the sin squared term is 0.9.

What pattern do you notice in your plot?

2.2 The Bloch Sphere

For systems with just two possible states (like two neutrino flavors), we can visualize superposition using something called the **Bloch sphere**.

What is it? The Bloch sphere is like a globe where every point on the surface represents a possible quantum state:

- The North Pole represents being 100 % in one state (say, $|\nu_e\rangle$)
- The South Pole represents being 100 % in the other state ($|\nu_\mu\rangle$)
- Every other point on the sphere represents a superposition of both states!

Reading the Sphere:

- Points near the North Pole: high probability of $|\nu_e\rangle$, low probability of $|\nu_\mu\rangle$

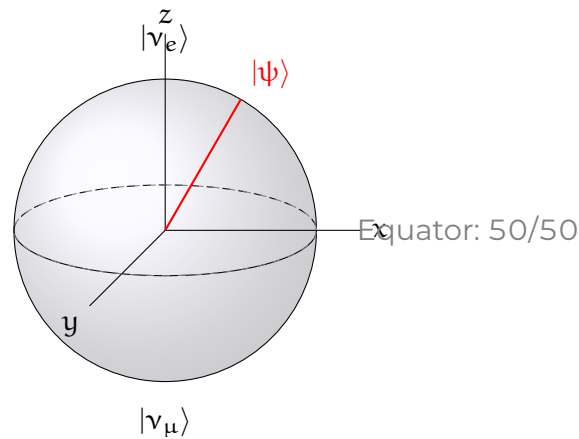


Figura 2: The Bloch Sphere - Every point represents a quantum state

- Points near the South Pole: low probability of $|\nu_e\rangle$, high probability of $|\nu_\mu\rangle$
 - Points on the equator: equal 50/50 probability of both states
- 4.** If a quantum state is at the North Pole of the Bloch sphere, what are the probabilities of measuring each state? What about at the South Pole?
- 5.** The fact that we observe neutrino oscillations at all tells us that neutrinos must have mass! Why? Look at the oscillation equation- What would happen to the probability if $\Delta m^2 = 0$? Calculate $P(\nu_e \rightarrow \nu_\mu)$ in this case.
- 5.** Look back at your plot from Question 4. As the neutrino travels, the probability oscillates up and down. Describe what you think is happening to the neutrino's position on the Bloch sphere as it travels through space. Draw the Bloch sphere at different positions along the curve.